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Research and Applications

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Leveraging open-source large language models for clinical information extraction in resource-constrained settings

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Abstract

Objective: We aimed to evaluate the zero-shot performance of open-source generative large language models (LLMs) on clinical information extraction from Dutch medical reports using the Diagnostic Report Analysis: General Optimization of NLP (DRAGON) benchmark.

Methods: We developed and released the `llm_extractinator` framework, a scalable, open-source tool for automating information extraction from clinical texts using LLMs. We evaluated 9 multilingual open-source LLMs across 28 tasks in the DRAGON benchmark, covering classification, regression, and named entity recognition (NER). All tasks were performed in a zero-shot setting. Model performance was quantified using task-specific metrics and aggregated into a DRAGON utility score. Additionally, we investigated the effect of in-context translation to English.

Results: Llama-3.3-70B achieved the highest utility score (0.760), followed by Phi-4-14B (0.751), Qwen-2.5-14B (0.748), and DeepSeek-R1-14B (0.744). These models outperformed or matched a fine-tuned RoBERTa baseline on 17 of 28 tasks, particularly in regression and structured classification. NER performance was consistently low across all models. Translation to English consistently reduced performance.

Discussion: Generative LLMs demonstrated strong zero-shot capabilities on clinical natural language processing tasks involving structured inference. Models around 14B parameters performed well overall, with Llama-3.3-70B leading but at high computational cost. Generative models excelled in regression tasks, but were hindered by token-level output formats for NER. Translation to English reduced performance, emphasizing the need for native language support.

Conclusion: Open-source generative LLMs provide a viable zero-shot alternative for clinical information extraction from Dutch medical texts, particularly in low-resource and multilingual settings.

Lay Summary

Our study tested the ability of open-source artificial intelligence (AI) language models to understand and extract information from Dutch medical reports without any prior training on the tasks. Using our custom `llm_extractinator` tool, we evaluated 9 models on 28 clinical tasks in the DRAGON 2024 benchmark. Several models around 14 billion parameters performed competitively, coming close to matching a top-performing, fine-tuned system built specifically for these tasks. Surprisingly, even without training, the generative models often outperformed the specialized model on several tasks, especially those involving numbers and structured reasoning. However, all models struggled with tasks that required identifying specific medical terms, likely due to the format required by the challenge. We also found that translating Dutch reports into English before analysis made results worse. Overall, our results suggest that modern generative AI can support clinical data analysis in Dutch with minimal setup, offering a flexible alternative to traditional models that require large amounts of labeled training data.

Key words: natural language processing; large language models; information storage and retrieval; artificial intelligence.

Introduction

Medical reports contain highly detailed patient information, including diagnoses, procedures, medications, and clinical observations, making them a valuable resource for data analysis for large-scale medical research.[1] This density of clinically relevant information is especially valuable for developing artificial intelligence (AI) applications in healthcare, which depend on large, well-labeled datasets.[2] When processed effectively, these reports can yield a wide variety of training labels, supporting the development of accurate and generalizable AI models.

The utility of medical reports is often limited by their unstructured textual format, which can vary significantly across institutions and individual practitioners.[3] Combined with the frequent use of domain-specific medical jargon, this lack of standardization presents a major challenge for information extraction, a critical step in converting raw clinical narratives into structured, machine-readable data.

Traditionally, the field of natural language processing (NLP) has relied on rule-based systems for information extraction, though these methods tend to struggle greatly with unstructured

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text.[4] The emergence of transformer-based models, such as Bidirectional Encoder Representations from Transformers (BERT),[5] enabled the extraction and structuring of meaningful data from more complex text. Domain-specific adaptations like Med-BERT[6] have further refined these capabilities, achieving state-of-the-art performance in tasks like text classification and extraction. However, their effectiveness hinges on the availability of large quantities of labeled training data, which limits their scalability and adaptability for new tasks.

Recent advancements in generative Large Language Models (LLMs) have introduced a transformative shift in NLP. These models can be adapted to diverse tasks through the use of prompting techniques, reducing or even eliminating the reliance on task-specific training data. Their application in healthcare has already shown promise in areas such as clinical decision support,[7–10] medical text summarization,[11] and question answering.[12,13]

However, a substantial portion of the current literature[7–9,12–20] is focused primarily on proprietary models such as OpenAI's GPT-4,[21] which pose challenges related to transparency, reproducibility, and ethical concerns in clinical applications. These systems generally require transmitting data via an API to external servers where the models are hosted. This approach raises significant concerns under modern privacy regulations governing medical data, which mandate strict oversight over any information leaving hospital IT systems. Additionally, the training of many proprietary models on mostly undisclosed datasets raises ethical questions about data sourcing and contamination, privacy, and representativeness.[22]

To address these limitations, the development and application of open-source LLMs have gained significant attention. These models offer researchers the opportunity to evaluate and adapt LLMs for medical tasks while ensuring greater accountability and control over input data by maintaining operations within local infrastructure. Open-source models also generally provide greater transparency regarding their pretraining datasets, enabling a clearer understanding of their limitations and biases.[23]

One of such limitations is the ability to effectively handle mid- to low-resource languages. Medical reports are predominantly written in the primary language of the care facility where they are produced. Proprietary models like GPT-4 benefit from extensive pre-training on datasets obtained through large-scale web scraping, which often include a variety of languages. In contrast, open-source LLMs are typically pretrained on more curated datasets, leading to a disproportionate representation of high-resource languages such as English, Chinese, and Spanish. This imbalance results in a significant performance gap between these widely spoken languages and less common ones.[24] The challenge is further compounded when dealing with text rich in specialized jargon, such as the terminology found in medical contexts, where the disparity in linguistic resources becomes even more pronounced. Despite the practical significance of these issues, research on the performance of open-source models in such contexts remains limited.[25–27]

The introduction of the Diagnostic Report Analysis: General Optimization of NLP (DRAGON) challenge[28] provides a valuable benchmark for addressing this issue. DRAGON includes 28 824 annotated medical reports from 5 care centers, covering 28 medically relevant information extraction tasks, such as classification, regression, and named entity recognition (NER), in the relatively uncommon Dutch language.

In this work, we present a systematic evaluation of several widely used open-source LLMs on domain-specific, resource-constrained language texts, with a focus on medical information extraction tasks. Our objective is to build a knowledge base identifying which models are most suitable for specific tasks in this setting, highlighting their strengths and limitations across various applications.

To support this evaluation, we developed a user-friendly and scalable framework that automates the application of open-source LLMs to diverse information extraction tasks on medical datasets in a language-agnostic manner. The framework enforces structured JavaScript Object Notation (JSON) output generation, enabling standardized and machine-readable outputs that facilitate both seamless evaluation and integration into downstream clinical or analytical pipelines. This design lowers the barrier to entry for deploying such models in complex, domain-specific contexts, while ensuring consistency and usability of the extracted information. Furthermore, the framework includes a web-based user interface that eliminates the need for any coding expertise, allowing less technical researchers to easily configure tasks, define output formats, and run models through an intuitive point-and-click environment.

The main contributions of our work are as follows:

1) We introduce and publicly release `llm_extractinator`, a scalable, language-agnostic, open-source framework for automating data extraction tasks with LLMs, designed for ease of use and broad applicability. It is available at https://github.com/DIAGNijmegen/llm_extractinator.
2) We perform a comprehensive evaluation of 9 widely used open-source LLMs on 28 medically relevant information extraction tasks using Dutch clinical reports in a zero-shot setting, as visually summarized in Figure 1. This evaluation offers a realistic estimate for model performance and practical insights into the utility of generative LLMs in resource-constrained, domain-specific environments. All associated code repositories are available in Note S5.

By focusing on smaller, open-source generative models, our work contributes to bridging the gap between state-of-the-art AI capabilities and practical, real-world applications in healthcare. This study not only fills a critical void in the literature but also lays the foundation for future research in leveraging open-source LLMs for multilingual and resource-constrained medical environments.

Methods

llm_extractinator

`llm_extractinator` (https://github.com/DIAGNijmegen/llm_extractinator) is an open-source pipeline that converts free text into structured JSON outputs based on user-defined schemas. Figure 2 gives an overview; below we describe the 4 stages of every run.

Task specification

A *Taskfile* (JSON) defines the data source, task description, the field to parse, and the desired output schema, which can be either Pydantic or pure JSON. Taskfiles can be created using the integrated Streamlit Studio GUI, offering a drag-and-drop interface with live schema previews, or by directly editing the JSON and parser script. Additionally, a `schema-`

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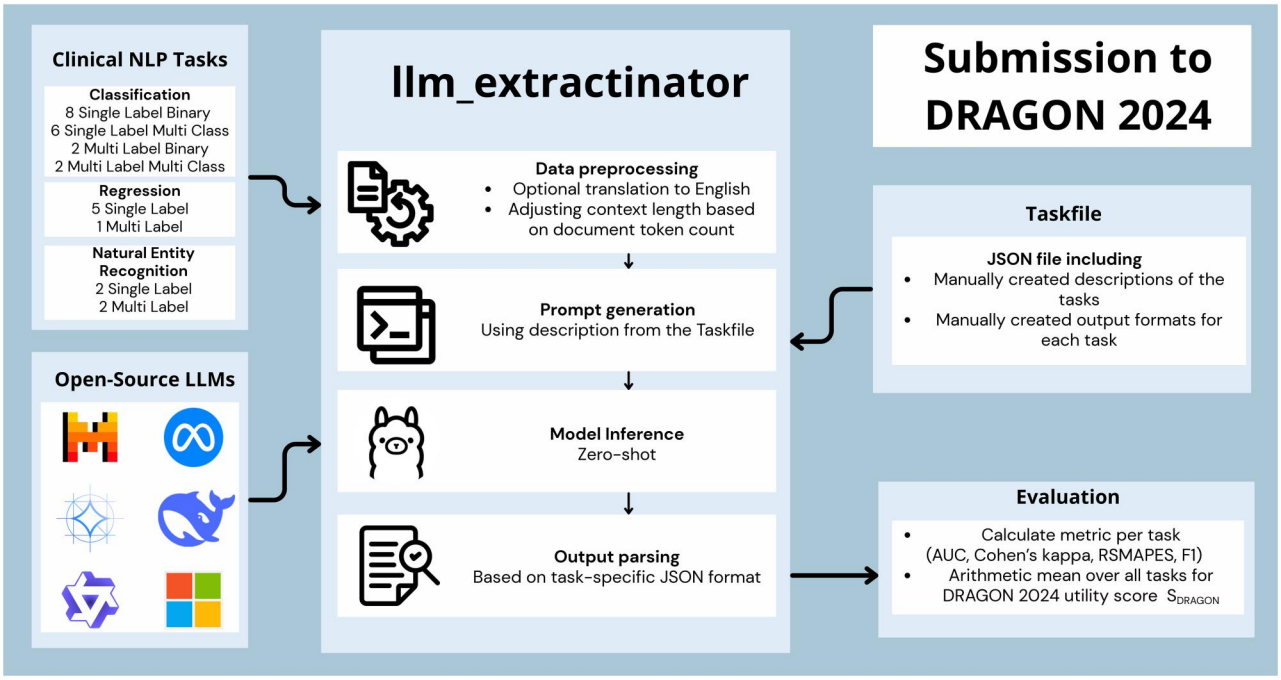

Figure 1. Overview of our submissions to the DRAGON 2024 challenge using the llm_extractinator framework. We evaluate 9 distinct LLMs: Mistral-Nemo-12B,[29] Llama-3.1-8B, Llama-3.2-3B, and Llama-3.3-70B,[30] Gemma-2-2B and Gemma-2-9B,[31] Phi-4-14B,[32] Qwen-2.5-14B,[33] and DeepSeek-R1-14B.[34] The input token length of each of the 28 clinical NLP tasks is measured, and model context windows are adapted accordingly. In 3 experiments, input text is translated to English using the LLM itself. For each task, we define a description and expected output format in a JSON-based Taskfile. This metadata guides prompt generation, which is followed by model inference and automatic output parsing. Task performance is assessed using the appropriate metric (AUC, Cohen's kappa, RSMAPE, or F1), and the final DRAGON 2024 utility score S_{DRAGON} is computed as the arithmetic mean across all task metrics.

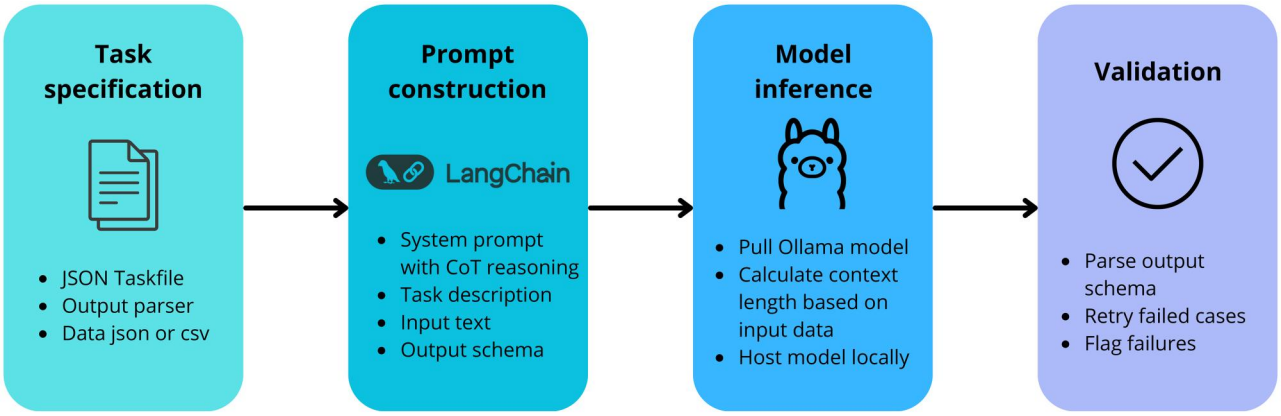

Figure 2. Overview of the `llm_extractinator` pipeline. The framework consists of 4 modular stages. (1) **Task specification**: users define the input source, task description, and output schema. (2) **Prompt construction**: the framework leverages LangChain to build a prompt that combines task instructions, the output schema, and zero-shot chain-of-thought reasoning cues. (3) **Model inference**: the prompt is sent to an LLM served through Ollama, which handles local deployment and manages resource-efficient inference through context-length scaling. (4) **Post-processing and validation**: the LLM output is parsed into the user-defined schema, with automatic reformatting of invalid responses and fallback placeholder injection to ensure pipeline continuity. Each stage is configurable but designed to operate out-of-the-box with default settings.

`builder` tool is included to assist in creating a Pydantic parser without any necessary coding knowledge.

Prompt construction

At inference time, LangChain[35] dynamically constructs a prompt by combining the raw input text, output schema, and task description. The prompt includes a general system message with zero-shot chain-of-thought reasoning instructions,[36] enforced through an explicit `reasoning` field in the output schema. For non-English input, optional automatic translation can be applied prior to extraction.

Model inference

`llm_extractinator` supports any LLM hosted on the Ollama hub.[37] The framework automatically downloads the selected model and runs it locally with user-defined inference settings. It dynamically determines the appropriate context length based on input size. For improved efficiency, inputs

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can optionally be split into short and long subsets, each processed with a tailored context window.

Post-processing and validation

Model outputs are validated against the defined schema using Ollama's enforced formatting. If validation fails, the framework initiates up to 3 self-correction cycles, in which the model receives its previous response and a list of schema violations. Remaining failures are replaced with empty or random placeholders to prevent downstream pipeline interruptions. These cases can later be flagged for manual review, though this functionality was disabled in the current study, where evaluation was performed fully automatically.

DRAGON challenge

To evaluate model performance, we applied our framework to the DRAGON challenge,[28] a benchmark initiative for clinical NLP tasks in Dutch. The DRAGON dataset comprises 28 824 annotated medical reports collected from 5 Dutch healthcare institutions, covering 28 clinically relevant tasks. These tasks span a diverse range of categories: 8 single-label binary classification tasks, 6 single-label multi-class classification tasks, 2 multilabel binary classification tasks, 2 multilabel multi-class classification tasks, 5 single-label regression tasks, 1 multilabel regression task, 2 single-label NER tasks, and 2 multilabel NER tasks. A complete overview of the tasks is provided in Figure 3 and in Note S1. A representative input text for each tasks is available at: https://github.com/DIAG-Nijmegen/dragon_sample_reports.

The challenge is hosted on the Grand Challenge platform,[38] a fully cloud-based environment powered by Amazon Web Services. The full challenge workflow involves 2 stages: fine-tuning BERT-like models on a provided training and validation dataset, followed by inference on a separate test set. However, since our study aims to evaluate the zero-shot capabilities of generative LLMs, we bypassed the fine-tuning phase and conducted direct inference via prompting.

The outputs generated by the models were post-processed into the JSON format required for automatic evaluation. These predictions were then evaluated against the ground truth test labels, producing task-specific performance scores. For the binary classification tasks (T1-T8), performance was assessed using the Area Under the Receiver Operating Characteristic Curve (AUC). For the multi-class classification tasks (T9-T14), performance was measured using either unweighted or linearly weighted Cohen's Kappa, depending on the task. Multi-label classification tasks (T15-T18) were evaluated using either macro-averaged AUC or unweighted Kappa. Regression tasks (T19-T24) used the Robust Symmetric Mean Absolute Percentage Error Score (RSMAPES) with task-specific tolerance margins. Named entity recognition tasks (T25-T28) were evaluated using macro or weighted F1 scores. All metrics were computed automatically by the Grand Challenge platform upon submission, and we did not have direct access to any per-example results.

The individual task scores were aggregated into the DRAGON 2024 utility score S_{DRAGON}, computed as the arithmetic mean of the performance metrics across all 28 tasks. While the standard DRAGON evaluation protocol recommends five test runs using different random seeds to account for sampling variability, we opted for a single test run by setting the model's sampling temperature to 0. This configuration enforces fully deterministic outputs in token generation, allowing for more stable, direct comparisons of zero-shot model performance. This approach also offers practical advantages by substantially reducing computational costs, as it eliminates the need for 4 additional inference runs per model.

Models

We evaluated 9 widely used open-source multilingual LLMs available through the Ollama model hub: Llama3.1-8B, Llama3.2-3B, and Llama3.3-70B,[30] Gemma2-2B and Gemma2-9B,[31] Phi4-14B,[32] Qwen2.5-14B,[33] DeepSeek-R1-14B,[34] and Mistral-NeMo.[29] For inference efficiency, all models were run in 4-bit quantized format. Specifically, we used the q4_0 quantization scheme for Mistral-Nemo and Gemma2, and q4_K_M for the remaining models. These configurations reflect the default quantization settings provided by the Ollama model hub.

Our primary focus was on models with fewer than 15 billion parameters to ensure practical feasibility within typical hospital IT environments. All such models can be run on consumer-grade GPUs with 12GB of VRAM when quantized to 4-bit precision. Although the llm_extractinator framework supports CPU offloading to accommodate larger models on limited hardware, this results in significant reductions in processing speed, rendering it impractical for clinical deployment. Given that most healthcare facilities lack ready access to high-performance GPUs, we prioritized smaller models for this study. Nevertheless, we also included one larger model (Llama-3.3-70B) to serve as a high-performance benchmark for institutions with greater computational resources. This helps contextualize the performance of smaller models and provides a reference point for future hardware scaling scenarios.

Prompting strategies

All experiments were conducted under a strict zero-shot setting, meaning that no task-specific fine-tuning was employed and no examples were provided in-context. This approach tests the models' inherent capabilities to perform unfamiliar medical tasks based solely on their pretrained knowledge. To encourage model reasoning, we consistently applied zero-shot chain-of-thought prompting across all tasks. A full list of all prompts used is provided in Note S4.

Moreover, we investigated the effect of translating the original Dutch reports into English prior to inference, given that the LLMs were predominantly trained on English corpora. This intermediate translation step was hypothesized to improve performance by aligning the input language with the models' primary training distribution, thus potentially reducing comprehension errors arising from linguistic mismatch.

Statistical analysis

Task-level scores with and without in-context translation were compared for each model using the non-parametric Kruskal-Wallis H-test. *P*-values were Holm-Bonferroni adjusted ($\alpha = 0.05$), and effect sizes reported as matched-pairs rank-biserial correlations r. Statistical analyses were performed using Python 3.11 with SciPy v1.15.3.

Results

We evaluated 9 publicly available generative LLMs on the 28 tasks of the DRAGON challenge using the `llm_extractinator` framework under zero-shot conditions. A representative

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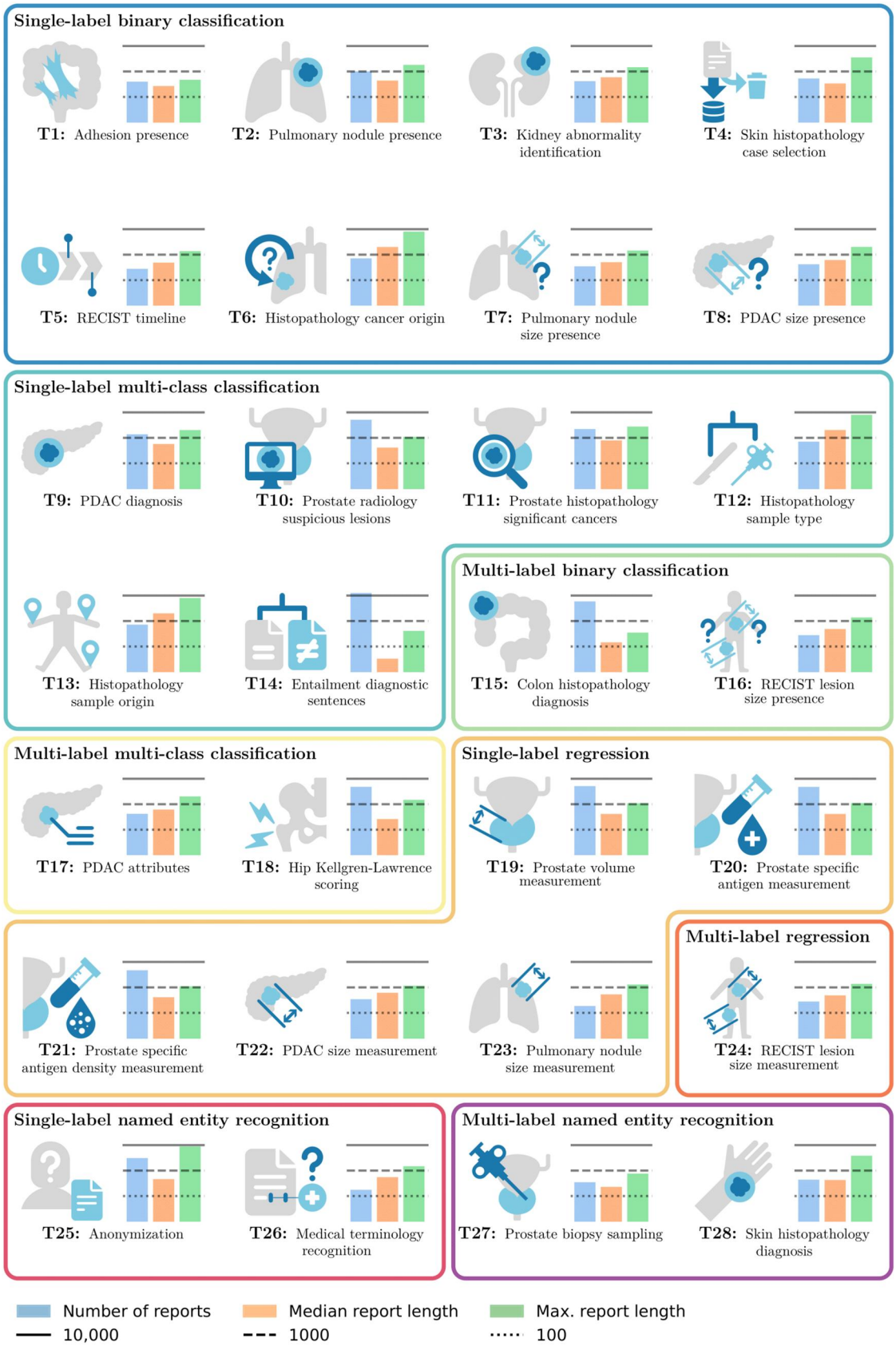

Figure 3. An overview of the 28 different tasks of the DRAGON challenge grouped by task type. The bar graphs show the number of reports, the median report length, and the maximum report length in each dataset based on tokenization using an xlm-roberta-base tokenizer. Figure reproduced with permission from Bosma et al.[28]

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input text for each tasks is available at: https://github.com/DIAGNijmegen/dragon_sample_reports.

We followed the proposed metrics of the challenge organizers to quantify performance: area under the receiver-operating-characteristic curve (AUC) for binary classification, Cohen's κ for multi-class classification, RSMAPES for regression, and F_1 score for NER. To facilitate model-level comparisons, we utilize the DRAGON 2024 utility score, S_{DRAGON}, defined as the arithmetic mean of each model's performance across all 28 tasks. The resulting score lies in the range [0,1], with 1 indicating perfect performance.

Additionally, the challenge organizers provide interpretability thresholds per metric which we use to categorize each result into one of 6 qualitative performance tiers: *Excellent*, *Good*, *Moderate*, *Poor*, *Minimal*, or *Fail*. Additional details on the metrics can be found in Note S2. The full results are documented in Note S3.

Model-level performance

Model performance naturally clustered into 3 general tiers. The top-performing group consisted of Llama-3.3-70B ($S_{\text{DRAGON}} = 0.760$), Phi-4-14B (0.751), Qwen-2.5-14B (0.748), and DeepSeek-R1-14B (0.744). Llama-3.3-70B scores best with an *Excellent* performance on 12 out of 28 tasks. Phi-4-14B achieved this performance on 10 out of 28 tasks, followed closely by Qwen-2.5-14B and DeepSeek-R1-14B, each with 9.

A second tier included Gemma2-9B and Mistral-Nemo-12B, both achieving $S_{\text{DRAGON}} = 0.688$, with *Good* or better performance on roughly half the tasks. Llama-3.1-8B scored notably lower ($S_{\text{DRAGON}} = 0.588$), achieving *Excellent* or *Good* performance on just 7 tasks.

The lowest tier comprised Llama-3.2-3B ($S_{\text{DRAGON}} = 0.271$), which achieved only *Minimal* to *Fail* performance across all tasks. Gemma2-2B consistently failed to produce valid JSON outputs and thus could not be evaluated meaningfully.

Table 1 summarizes S_{DRAGON} scores alongside the number of tasks each model performed at each qualitative level. RoBERTa large with domain-specific pretraining, the best performing baseline model provided by the challenge organizers, is included for reference. Figure 4 visualizes average performance per task type across all models we tested. While the tiered structure is generally consistent across task types, certain models exhibit domain-specific strengths. For instance, Mistral-Nemo-12B performed comparably to top-tier models on regression tasks but underperformed on multilabel classification. Conversely, Gemma2-9B demonstrated relatively weaker performance on regression despite competitive results in other task types.

Task-level performance

Figure 5 shows task-specific performance distributions for the top 4 models. Across the 6 regression tasks, all models achieved scores ≥ 0.87, with an average RSMAPES of 0.971 and 22 out of 24 model-task combinations rated *Excellent*. Binary classification tasks showed greater variability: while the group mean AUC of the 4 models over all tasks was 0.84, certain tasks (eg, T04 and T06) saw performance near chance level for at least one model.

Ordinal classification tasks revealed broad score distributions, with Cohen's κ values ranging from 0.51 to 0.98. The largest intra-task spread (T14, $\sigma = 0.09$) illustrates the potential impact of model selection. Some tasks (eg, T10 and T12) consistently produced high scores with low inter-model variability. In contrast, tasks T11, T14, and T18 displayed high variance and low scores, indicating task-level difficulty or sensitivity to model architecture.

NER performance was uniformly poor: none of the evaluated models exceeded an F_1 score of 0.47. The modal qualitative label for NER tasks was *Fail*.

Comparison to BERT-style baseline model

Table 2 provides a detailed task-by-task comparison between the current top-performing model in the DRAGON 2024 challenge, DRAGON RoBERTa Large Domain-specific (https://grand-challenge.org/algorithms/dragon-roberta-large-domain-specific/), and our best performing model Llama-3.3 (https://grand-challenge.org/algorithms/llm-extractinator-llama33/). The better-performing model for each task is highlighted in bold. RoBERTa's results are reported as the mean and SD from 5-fold cross-validation, where the Llama-3.3 scores are derived from a single deterministic inference run with zero temperature.

The RoBERTa model achieved a higher overall DRAGON 2024 utility score ($S_{\text{DRAGON}} = 0.819 \pm 0.021$) than any of the generative LLMs. However, across the 28 tasks, DRAGON RoBERTa Large Domain-specific achieved higher scores than Llama-3.3 on only 11 tasks (T01, T02, T04, T05, T09, T11, T15, T25, T26, T27, T28), with performance differences exceeding the upper bound of RoBERTa's SD, whereas the Llama model scored higher than RoBERTa on 14 tasks (T06, T07, T08, T12, T13, T16, T17, T18, T19, T20, T21, T22, T23, T24). For the remaining 3 tasks (T03, T10, and T14), the score of the generative LLM fell within one SD of RoBERTa's mean score.

Table 1. DRAGON 2024 utility scores (S_{DRAGON}) and qualitative ratings across the 28 DRAGON tasks.

Model	S_{DRAGON}	Excellent	Good	Moderate	Poor	Minimal	Fail
LLaMA 3.3 70B	0.760	12	3	7	3	0	3
Phi-4 14B	0.751	10	6	5	4	0	3
Qwen 2.5 14B	0.748	9	7	6	2	1	3
DeepSeek-R1 14B	0.744	9	6	5	5	1	2
Gemma 2 9B	0.688	6	7	6	4	1	4
Mistral-Nemo 12B	0.688	7	6	5	5	2	3
LLaMA 3.1 8B	0.588	3	4	4	4	5	8
LLaMA 3.2 3B	0.271	0	0	0	0	7	21
RoBERTa Large	0.819	10	8	6	2	2	0

RoBERTa large is included as a reference, representing the current best-performing BERT-style baseline model provided by the challenge organizers (https://grand-challenge.org/algorithms/dragon-roberta-large-domain-specific/). Unlike our models, this baseline was trained directly on all 28 tasks and is not evaluated in a zero-shot setting.

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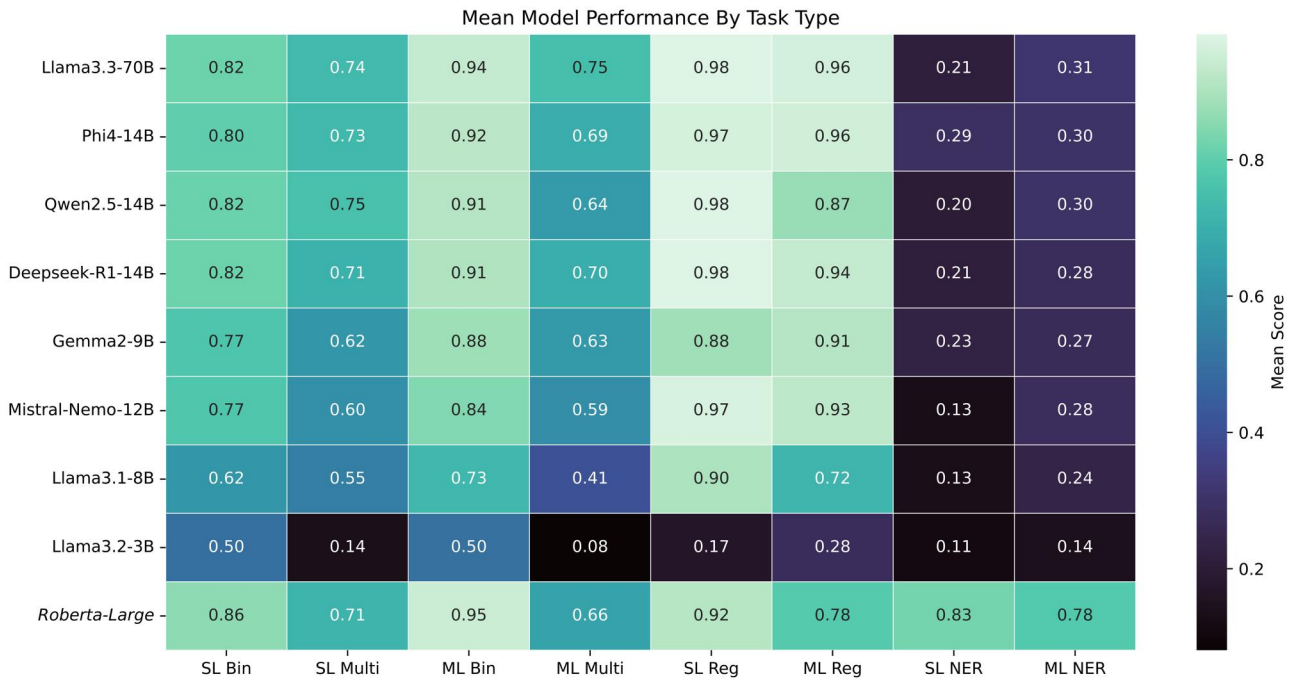

Figure 4. Heatmap illustrating the average performance of models across various task categories. Each cell shows the mean model score across tasks within a category. Scores range from 0 (worst) to 1 (best), except for multiclass classification tasks evaluated with Cohen's kappa, which ranges from −1 (complete disagreement) to 1 (perfect agreement), with 0 indicating chance. For binary classification, 0.5 reflects chance-level performance. Task types are abbreviated as follows: **SL** = *Single-label*, **ML** = *Multi-label*, **Bin** = *Binary classification*, **Multi** = *Multi-class classification*, **Reg** = *Regression*, **NER** = *Named Entity Recognition*. The colormap represents average performance scores, with exact values annotated in each cell. The evaluation metric varies by task type: AUC is used for binary classification, Cohen's Kappa for multi-class classification, RSMAPES for regression, and F1 score for NER. The performance of the best performing baseline RoBERTa model of the challenge organizers (https://grand-challenge.org/algorithms/dragon-roberta-large-domain-specific/) is provided for reference.

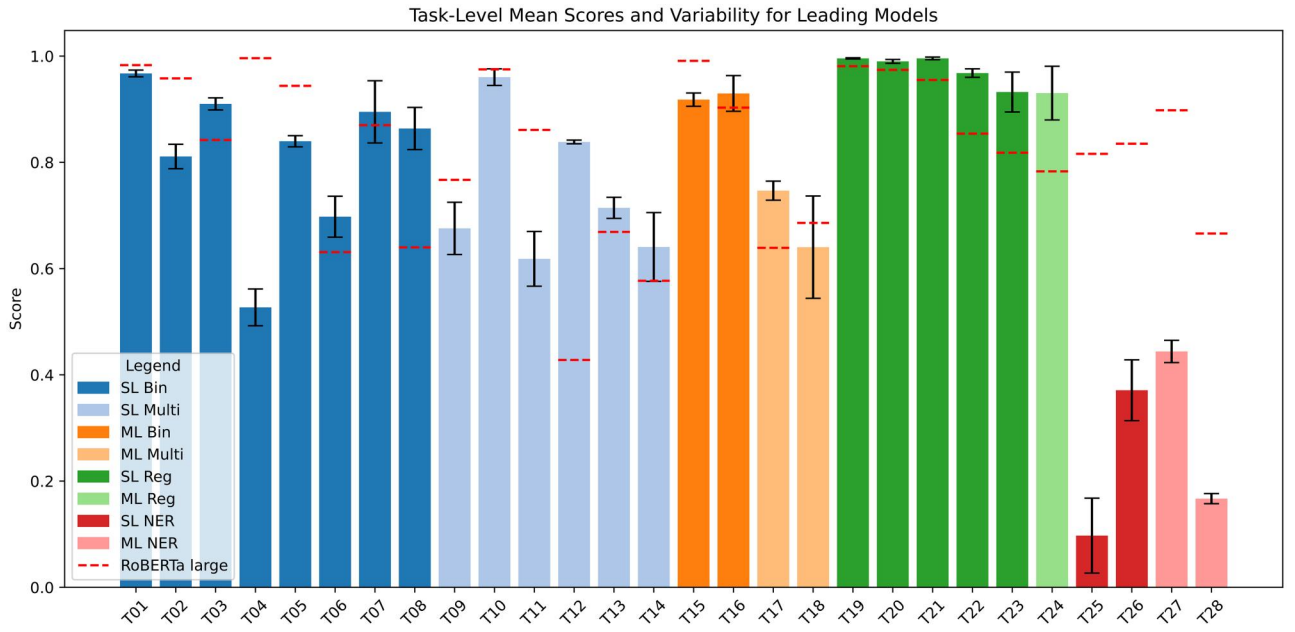

Figure 5. Mean performance scores and SD for each of the 28 tasks, computed over the final evaluation metric across the top 4 performing models (Llama-3.3-70B, Phi-4-14B, Qwen-2.5-14B, and DeepSeek-R1-14B). Task types are color-coded and are abbreviated as follows: **SL** = *Single-label*, **ML** = *Multi-label*, **Bin** = *Binary classification*, **Multi** = *Multi-class classification*, **Reg** = *Regression*, **NER** = *Named Entity Recognition*. The mean performance per task of the best performing baseline RoBERTa model of the challenge organizers (https://grand-challenge.org/algorithms/dragon-roberta-large-domain-specific/) is provided as dotted red lines for reference.

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Table 2. Task-wise comparison between LLaMA3.3-70B (https://grand-challenge.org/algorithms/llm-extractinator-llama33/) and DRAGON RoBERTa large domain-specific (https://grand-challenge.org/algorithms/dragon-roberta-large-domain-specific/).

Task	LLaMA3.3	RoBERTa large
T01	0.971	**0.983** ± 0.004
T02	0.788	**0.958** ± 0.008
T03	0.923	0.842 ± 0.096
T04	0.500	**0.996** ± 0.001
T05	0.840	**0.944** ± 0.010
T06	**0.708**	0.631 ± 0.042
T07	**0.955**	0.870 ± 0.040
T08	**0.883**	0.640 ± 0.050
T09	0.619	**0.767** ± 0.039
T10	0.978	0.975 ± 0.003
T11	0.669	**0.861** ± 0.003
T12	**0.842**	0.428 ± 0.057
T13	**0.736**	0.669 ± 0.022
T14	0.573	0.577 ± 0.009
T15	0.917	**0.991** ± 0.010
T16	**0.959**	0.903 ± 0.015
T17	**0.767**	0.639 ± 0.074
T18	**0.732**	0.686 ± 0.015
T19	**0.995**	0.981 ± 0.002
T20	**0.991**	0.974 ± 0.002
T21	**0.993**	0.955 ± 0.012
T22	**0.976**	0.854 ± 0.003
T23	**0.955**	0.818 ± 0.012
T24	**0.961**	0.783 ± 0.003
T25	0.028	**0.816** ± 0.007
T26	0.401	**0.835** ± 0.003
T27	0.467	**0.898** ± 0.010
T28	0.161	**0.666** ± 0.035
Score	0.760	**0.819** ± 0.021

RoBERTa scores include SD based on 5-fold cross-validation. A score is bolded when it is higher than the other model's and lies outside of RoBERTa's SD.

Effect of in-context translation

To assess the impact of in-context translation, we compared model performance on Dutch inputs with and without prior translation to English by the LLMs themselves. To assess the impact of in-context translation, we compared model performance on Dutch inputs with and without prior translation to English by the LLMs themselves. This translation strategy led to statistically significant performance degradation for Phi-4-12B (S_{DRAGON} 0.751 → 0.533, $\Delta = -0.22$, $H = 11.56$, $P_{\text{adj}} < .001$, $r = -0.94$) and Llama-3.1-8B (0.588 → 0.337, $\Delta = -0.25$, $H = 11.56$, $P_{\text{adj}} < .001$, $r = -0.94$). For Mistral-Nemo-12B, the decrease from 0.688 to 0.573 ($\Delta = -0.11$, $H = 2.50$, $P_{\text{adj}} = .114$, $r = -0.30$) was not statistically significant after adjustment. These findings indicate that in-context translation can markedly impair downstream performance for some models.

Figure 6 details the differences in performance for all models where translation was tested, showing consistent task-level reductions in the relevant performance metric. These results suggest that translation-induced noise undermines clinical information extraction accuracy, underscoring the importance of native-language inference for domain-specific tasks.

Discussion

In this study, we evaluated the zero-shot performance of 9 widely used open-source LLMs using the llm_extractinator framework on the DRAGON challenge, a Dutch clinical NLP benchmark. Our results highlight both the promise and limitations of deploying such models for real-world information extraction tasks in healthcare.

We found that models with around 14B parameters, including Phi-4, Qwen-2.5, and DeepSeek-R1, performed well across most tasks, achieving average DRAGON utility scores near 0.75. The Llama-3.3-70B model outperformed all others with a utility score of 0.76, consistent with prior findings that larger models tend to generalize better.[39,40] However, this improvement came with significant computational cost and only translated into higher task-level performance in 11 of 28 cases. This suggests that performance gains from scaling are not uniform across task types, and that deploying larger models is most justifiable when computational resources are readily available and the marginal performance gains are considered worthwhile.

Regression tasks, such as extracting lesion sizes or PSA levels, were a relative strength for all tested LLMs. These results contrast with the weaker performance by fine-tuned BERT-style models on the multi-label regression tasks in particular. Generative models appear to handle numeric value reproduction especially well due to their copy-and-reason capabilities. This aligns with prior intuitions that generative models retain quantitative tokens during inference whereas this is more difficult for encoder-based models.[41]

Performance declined markedly on classification tasks and collapsed on NER. Even the strongest models achieved F1 scores below 0.5 on the latter. This underperformance was likely exacerbated by the token-level output format required by the DRAGON challenge. Generative models are not naturally suited for generating sparsely populated token-level lists, and our structured prompting followed by post-processing likely introduced conversion errors. It has been shown in other work that more suitable evaluation formats can yield good performance on similar tasks.[42] As such, our results represent only a conservative estimate of model capabilities on these tasks.

Additionally, certain tasks were inherently unsuited to zero-shot evaluation and were therefore unlikely to succeed. For instance, Task 04 (Skin histopathology case selection) asked models to determine whether a pathology report should be excluded based on vague criteria such as being incomplete or lacking a definitive diagnosis. In the intended use case, where models are trained on labeled examples, such patterns could be learned. However, in a zero-shot setting, where no task-specific feedback or examples are available, the prompt alone provides insufficient guidance. This limitation is reflected in the near-random performance of even the top-performing models on this task.

Annotation quality also influenced model performance. As reported by the DRAGON organizers, some tasks showed relatively low inter-annotator agreement. Notably, Task 06 (histopathology cancer origin) had a Krippendorff's Alpha of 0.333, Task 14 (textual entailment) scored 0.550, Task 17 (PDAC attributes) 0.677, and Task 18 (hip osteoarthritis scoring) 0.557. Inconsistent labeling likely contributed to higher variance across models and limited overall accuracy, even for top-performing systems.

While the DRAGON leaderboard is currently led by fine-tuned encoder models, our zero-shot evaluation of generative models paints a more nuanced picture. Although Llama-3.3 trailed the top-performing RoBERTa-based model overall

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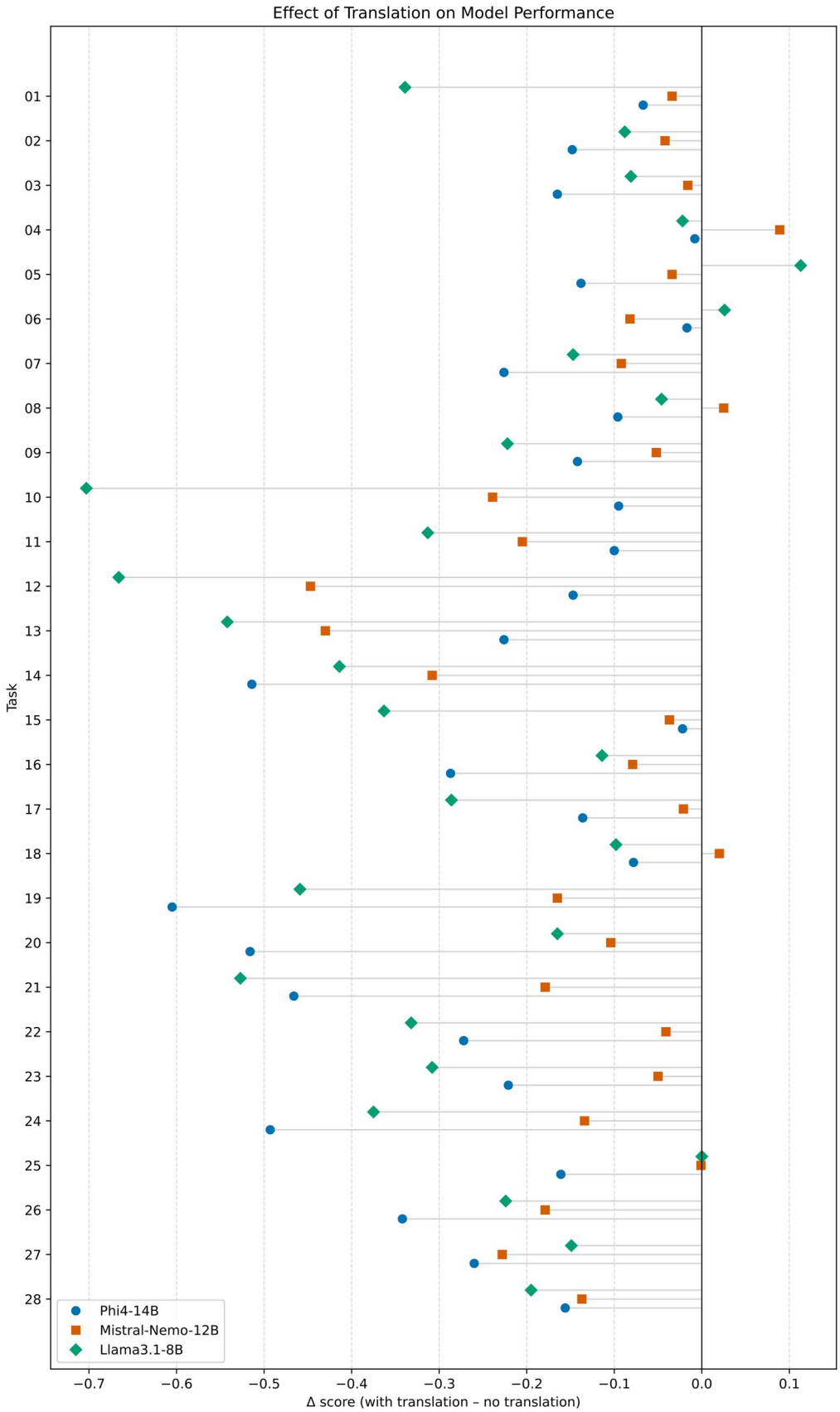

Figure 6. Impact of machine translation on LLM performance across 28 clinical NLP tasks in the DRAGON challenge. This plot illustrates the performance deltas (with translation—without translation) for Phi-4-14B, Mistral-NeMo-12B, and Llama-3.1-9B. Negative deltas indicate that translation on average degrades model performance across tasks.

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(0.760 vs 0.819 utility score), this difference is primarily due to strong relative performance of the RoBERTa model on NER tasks and Task 04. Excluding these tasks shifts the average score in favor of Llama-3.3. Its S_{DRAGON} rises to 0.858, while RoBERTa's drops to 0.814. This suggests complementary strengths: encoder models excel at token-level classification, while generative LLMs are better suited to structured inference and regression tasks.

Importantly, these comparisons must be contextualized within the operational and data constraints of real-world deployments. Fine-tuned RoBERTa models require supervised training on labeled data for each task, and are tightly coupled to their respective training distributions. By contrast, Llama-3.3 and other generative LLMs were evaluated strictly in a zero-shot setting, without any parameter updates or task-specific examples in-context. That they perform comparably under these conditions, sometimes even exceeding RoBERTa's performance, suggests that generative models are becoming increasingly viable alternatives for scalable, plug-and-play clinical NLP, especially in settings where labeled data is scarce or task requirements evolve frequently.

Another key insight from our analysis is the significant negative impact of translating Dutch medical texts into English prior to inference. Across the board, naïve translation consistently degraded performance. While the literature presents mixed findings depending on the context, with some studies reporting benefits from pre-translation[43,44] while others observe better outcomes without it,[45] our results suggest that translation on this dataset introduces artifacts and erodes clinical nuance. These findings underscore the need for robust native-language support in clinical NLP tools.

Smaller models, such as Llama-3.2-3B and Gemma-2-2B, consistently failed across tasks, producing nonsensical outputs. This establishes a practical lower bound for model scale for zero-shot clinical NLP in a non-English language. Our `llm_extractinator` framework supports efficient inference of larger models on consumer-grade GPUs. This effectively eliminates the need to rely on underpowered models. While smaller models may offer marginal improvements in inference speed, the trade-off in output quality is steep: the risk of generating entirely unusable results outweighs any computational performance gains.

This study has several limitations. First, a uniform prompting approach was used across tasks and models, without extensive task-specific engineering. While this supports reproducibility, it likely underestimates achievable performance. Second, our evaluation was limited to zero-shot settings in Dutch. Generalizability to other languages remains to be tested and there is room for future research to explore the effects of few-shot prompting,[46] lightweight instruction-tuning,[47] or retrieval-augmented generation[48] on model performance. Third, due to resource constraints, we only evaluated one model over 15B parameters and did not include any of the largest open-source LLMs. Future work should explore their capabilities, especially in high-resource settings. Finally, since all evaluations were handled through the Grand Challenge platform without access to per-example outputs, we were unable to perform detailed error analysis. This restricts insights into where and why models may have failed.

In summary, this work demonstrates that open-source generative LLMs can serve as powerful tools for medical information extraction in Dutch. With minimal infrastructure or labeled data, several models approach or surpass fine-tuned encoder baselines in clinical NLP tasks. By streamlining this process through our llm_extractinator framework, we lower the barrier to applying these models in real-world clinical research.

Author contributions

Luc Builtjes (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft), Joeran Bosma (Data curation, Resources, Writing—original draft), Mathias Prokop (Supervision), Bram van Ginneken (Supervision, Writing—original draft), and Alessa Hering (Conceptualization, Supervision, Writing—original draft)

Supplementary material

Supplementary material is available at *JAMIA Open* online.

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Conflicts of interest

The authors have no competing interests to declare.

Data availability

This study is based on data from the DRAGON challenge hosted on the Grand Challenge platform (https://dragon.grand-challenge.org/). The DRAGON challenge is a Type 3 challenge, meaning that neither the training nor test datasets are accessible to participants. Instead, algorithms are developed and submitted by participants to be executed on hidden datasets maintained by the organizers. As such, the data are not available for download or direct inspection. Researchers can participate in the challenge and evaluate their methods by submitting their algorithms through the challenge platform.

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